EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 9A

1. I bet Cal will win this year

As huge fans of the Big Game, you and your friend want to bet on whether Cal or Stanford will win this year. You want to predict this year's result by analyzing historical records. Therefore, you decide to model this as a binary classification problem and do PCA for dimension reduction on the data you collected. The "+1" class represents victories of Cal and "−1" represents victories of Stanford.

After some research, you obtained a data matrix $\mathbf{A} \in \mathbb{R}^{d\times n}$,

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \tag{1}$$

where each of the n columns $\mathbf{a}_i$ denotes a game and each of the d rows of $\mathbf{A}$ contains information of a possibly relevant factor of the games (weather, location, date, air quality, etc).

(a) Let the full SVD of $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$. You project your data along $\mathbf{u}_1$ and $\mathbf{u}_2$ (the first two principal components), and for comparison you also project your data along two randomly chosen directions $\mathbf{w}_1$ and $\mathbf{w}_2$ as well. You get the two pictures in Figure 1, but you forgot to label the axes. Of the two figures below, which one is the projection onto the principal components and which one is the projection onto the random directions? **Match axes (i), (ii), (iii), (iv) to $\mathbf{w}_1, \mathbf{w}_2, \mathbf{u}_1$, and $\mathbf{u}_2$, and justify your answer.** *Note that there may be multiple correct matchings; you only need to find and justify one of them.*

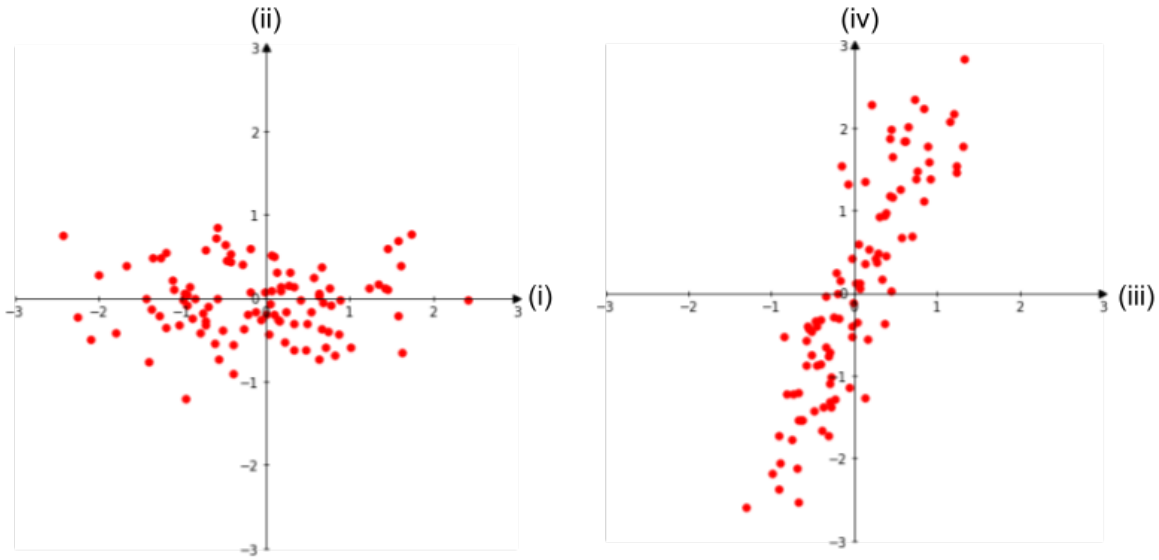


Figure 1: Projected datasets.

Answer: When we project data on the principal components, we require that one of the orthogonal axes maximizes the sum of squares of coordinates of the data points. We can see that the plot on the left is likely a projection onto the principal components, since the horizontal (i.e., x) axis maximizes the sum of the coordinates, which are the x-coordinates here. So we can deduce from this that (i) and

(ii) must correspond in some order to the principal components and (iii) and (iv) must correspond to the random directions. Since axis (i) has a larger sum of squares than axis (ii) in the diagram on the left, it must be the case that axis (i) corresponds to $\mathbf{u}_1$, and the other corresponds to $\mathbf{u}_2$. The data is centered at (0, 0), so we can equivalently compare the "spread" along any given axis.

i. $\mathbf{u}_1$ – This is the axis with the maximal spread of the data and therefore must correspond to the largest singular value. the single axis, across both plots, across which there is maximal spread of the data.
ii. $\mathbf{u}_2$ – the corresponding axis to $\mathbf{u}_2$ and also an axis for which the spread of the data is axis-aligned.
iii. $\mathbf{w}_1$ or $\mathbf{w}_2$ – seemingly a random projection. We don't know which one is $\mathbf{w}_1$ and $\mathbf{w}_2$ since they are random unit vectors and as such are independent of the data, so we can't tell from the plot.
iv. $\mathbf{w}_2$ or $\mathbf{w}_1$ – seemingly a random projection. We don't know which one is $\mathbf{w}_1$ and $\mathbf{w}_2$ since they are random unit vectors and as such are independent of the data, so we can't tell from the plot.

(b) In order to reduce the dimension of the data, we would like to project the data onto the first k principal components of $\mathbf{A}$, where k is less than the original data dimension d. **Show how to find the new vector $\mathbf{z}_i \in \mathbb{R}^k$ which is the k-dimensional, compressed version of $\mathbf{x}_i$.** You may use the SVD of $\mathbf{A}$.

Answer: Write the matrix $\mathbf{A}$ in terms of its SVD, namely $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$. In order to select the top k principal components, we can take the first k columns of $\mathbf{U}$, denote $\mathbf{U}_k$. To compress the data into k dimensions and find $\mathbf{z}_i$, we compute

$$\mathbf{z}_i = \mathbf{U}_k^\top \mathbf{x}_i \tag{2}$$

(c) Given a new set of projection coefficients denoted $\mathbf{z}_{\text{new}} \in \mathbb{R}^k$, we can define a classifier that will predict $+1$ (i.e., that Cal wins) if $\mathbf{w}_\star^\top \mathbf{z}_{\text{new}} > 0$ and -1 (i.e., that Stanford wins) otherwise.

Assume $d = 6$, $k = 4$, and $\mathbf{w}_\star = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}^\top$. Let $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$ for $\mathbf{A}$, and you find that $\mathbf{U}$ is given by the identity matrix, i.e. $\mathbf{U} = \mathbf{I}_d$. Now suppose the data point for this year's big game $\mathbf{x}_{2025} = \begin{bmatrix} 3 & 6 & 4 & 1 & 9 & 6 \end{bmatrix}^\top$. **Would you bet on Cal or Stanford to win? Justify your answer.**

(HINT: Don't forget to project your data onto the principal components.)

Answer: First, we need to project the new data point onto the k-dimensional subspace just like what we did to the original data

$$\mathbf{z}_{2025} = \mathbf{U}_k^\top \mathbf{x}_{2025} \tag{3}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}^\top \begin{bmatrix} 3 \\ 6 \\ 4 \\ 1 \\ 9 \\ 6 \end{bmatrix} \tag{4}$$

$$= \begin{bmatrix} 3 \\ 6 \\ 4 \\ 1 \end{bmatrix} \tag{5}$$

Then, we compute the classifier's predicted value

$$p_{2025} = \mathbf{w}_\star^\top \mathbf{z}_{2025} \tag{6}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 4 \\ 1 \end{bmatrix} \tag{7}$$

$$= 6 > 0 \tag{8}$$

Therefore, the classifier predicts the label for this data point to be "$+1$", thus you should bet for Cal to win this year!

2. Robotic Grasping with PCA (Adapted from EECS 16A Fa24 Exam 3)

As a robotics automation engineer, we would like to develop a robotic system that can pick up ellipse-shaped objects from a table, using only an overhead camera image as input.

In order to estimate the pose of our elliptical objects, we will characterize it by their center point (x, y), while their orientation is described by a unit vector pointing in the direction of their major axis(the longer axis).

To derive these positions, we will sample points in the image that appear to lie on the object. By doing so uniformly across the image, we assume that they are roughly representative of the object's actual position. An example plot is as follows:

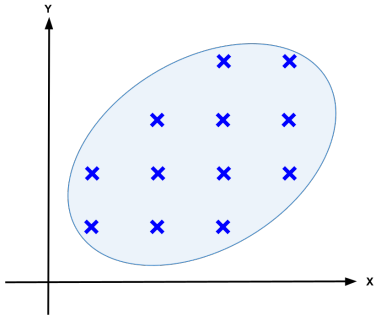


Figure 2: Example of Object Sampling

For one particular sampled object, assume that we store the (x, y) positions of our sampled points as **row data** in the following matrix:

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 1 & 2 \end{bmatrix}$$

(a) Given the data matrix $\mathbf{A}$, estimate the position $(\bar{x}, \bar{y})$ of the center of the ellipse.

Answer:

Given only the data in the matrix, the best estimate for the ellipse's center is simply the average of the samples. This can be computed by finding the vector averages of the columns of $\mathbf{A}$, or simply the average x and y coordinates of our points. Mathematically, we can see that

$$\bar{x} = \frac{3+1+2+3+1}{5} = \frac{10}{5} = 2$$

$$\bar{y} = \frac{2+1+2+3+2}{5} = \frac{10}{5} = 2$$

Therefore, the estimated center of our ellipse is $(2, 2)$.

(b) Ignoring the previous subpart, imagine that the data has now been centered about the origin, and can now be characterized by the following data matrix $\mathbf{A}'$.

$$\mathbf{A}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -2 & 2 \\ 2 & -2 \\ -1 & -1 \end{bmatrix}$$

Given this matrix, determine the best estimate for the unit vector $\hat{\mathbf{m}}$ that characterizes the direction of the ellipse's major axis. This axis should correspond to the **direction of maximum variance** in our sampled points.

Answer:

Since our data is stored in rows, the major axis vector should be our principal right singular vector, or the first row of our $\mathbf{V}^\top$ matrix in the SVD of $\mathbf{A}'$.

Rather than necessarily computing the complete SVD, we can instead simply compute $\mathbf{A}'^\top\mathbf{A}'$ and find the eigenvectors to construct $\mathbf{V}^\top$. Making sure that $\mathbf{V}^\top$ is orthonormal, the correct orientation vector $\hat{\mathbf{m}}$ will simply be the first row.

Mathematically, we find that

$$\mathbf{A}'^\top\mathbf{A}' = \begin{bmatrix} 1 & 0 & -2 & 2 & -1 \\ 1 & 0 & 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -2 & 2 \\ 2 & -2 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 10 & -6 \\ -6 & 10 \end{bmatrix}$$

By inspection, we can see that:

- $\begin{bmatrix} 10 & -6 \\ -6 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 16 \\ -16 \end{bmatrix} = 16 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- $\begin{bmatrix} 10 & -6 \\ -6 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

so the eigenvectors of $\mathbf{A}'^\top\mathbf{A}'$ are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ with eigenvalues 16 and 4 (this could also be discovered via the characteristic polynomial).

The vector with the largest eigenvalue (and thus largest singular value) is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Normalizing tells us that

$$\hat{\mathbf{m}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Now that we have successfully estimated the pose of our target object, we can actually pick it up use a gripper as shown below!

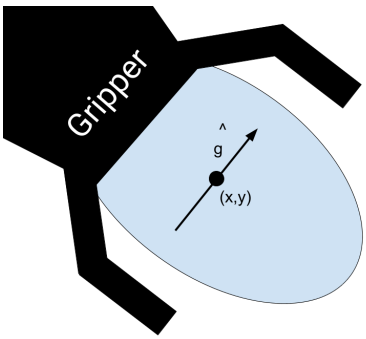


Figure 3: Grasp Visualization

(c) In order to maximize our chances of successfully grasping the object, however, we want to grasp along the **shortest** axis of our ellipse estimation. Assuming that the estimate of the dirrection of ellipse's major axis is $\hat{\mathbf{m}} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$, provide the unit vector $\hat{\mathbf{g}}$ along the **shortest** axis.

Answer:

Because we are only working in 2 dimensions, the shortest axis of our ellipse estimation must be orthogonal to the major axis of our ellipse, per the definition of the minor axis. There are multiple possible ways that students can derive the particular vector $\hat{\mathbf{g}}$ that corresponds to this direction:

- Observing that the provided $\hat{\mathbf{m}}$ vector is the same as their result from the previous subpart, we can observe that the shortest axis must be the secondary right principal vector, or the second row of the matrix $\mathbf{V}^\top$ that we derived. As a normalized vector, this automatically gives us

$$\hat{\mathbf{g}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

- Irrespective of the previous subpart, students can observe that the grasp axis should be a vector orthogonal to $\hat{\mathbf{m}}$, and then produce such a vector by inspection:

$$\left\langle \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0$$

  Thus,

$$\hat{\mathbf{g}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

- Alternatively to inspection, with an arbitrary identity basis vector (say $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$), we can extend an orthogonal basis with Gram-Schmidt to get

$$\mathbf{q} = \mathbf{e}_1 - \mathbf{proj}_{\hat{\mathbf{m}}} \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

  Normalizing, we see

$$\hat{\mathbf{g}} = \sqrt{2} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

All of the above gave us $\hat{\mathbf{g}} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$, but the vector $\hat{\mathbf{g}} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$ is also a valid solution.